Accelerating the Krusell–Smith Method

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October 2026

Abstract

This paper substantially accelerates the Krusell–Smith method, particularly the simulation stage in models with nontrivial market-clearing conditions. By precomputing agents' optimal responses to different prices, the method eliminates the need to repeatedly solve their decision problems during simulation. In the Bloom et al. (2018) model, the proposed grid-based method reduces simulation time from 290 minutes to 1.5 minutes and total computation time from approximately four days to 108 minutes. Because grid-based precomputation becomes costly as the number of states and equilibrium prices increases, I extend the approach using neural networks with a discrete–continuous policy representation that accommodates (S, s)-type adjustment rules. I then use the method to revisit aggregate comovement after a pure uncertainty shock in the Bloom et al. (2018) model. While the original specification fails to generate simultaneous declines in output, consumption, investment, and employment on impact, making the Frisch elasticity finite is sufficient to generate the missing comovement.

Keywords. Krusell–Smith method; Heterogeneous-agent models; Neural networks; Aggregate uncertainty; Nonconvex adjustment costs

JEL Classification. C63, C68, E22, E32

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I am indebted to my adviser Tatsuro Senga for his invaluable guidance and support. I am also grateful for comments and suggestions from Marlon Azinovic, Ji Huang, Hidehiko Matsumoto, Jacob Røpke, Simon Scheidegger, Takashi Shiono, Yucheng Yang, Jan Žemlička, and participants at conferences and workshops. This research was supported by a grant from the Koizumi and Ushioda Memorial Fund at Keio University. All remaining errors are mine. An earlier version of this paper circulated under the title "Accelerating the Krusell–Smith Algorithm with Deep Learning."

1 Introduction

Heterogeneous firm models are widely used for studying the macroeconomic implications of micro-level heterogeneity among plants and firms; however, analyzing cyclical dynamics using this class of models with aggregate uncertainty poses a significant computational challenge. As established by Terry (2017), the Krusell-Smith (KS) method is the superior method among alternatives—particularly for handling large shocks and non-linear dynamics—offering unmatched accuracy and flexibility for adapting to diverse macroeconomic heterogeneous agent models. This approach can capture complex phenomena like occasionally binding constraints, non-stationary dynamics, and multiple equilibria that simpler linearization methods miss. Ironically, this same generality makes it the most computationally resource-demanding method required to achieve such precision.[1]

This paper makes three contributions. First, I develop a grid-based method that substantially accelerates the KS method while preserving its basic structure and benchmark accuracy. In the Bloom et al. (2018) model, total computation time falls from 6,141 minutes, or approximately four days, to 108 minutes. Second, I extend the method using neural networks, named Neural Network-based Krusell-Smith Method (NN-KS), to accommodate larger models. Additionally, I construct a policy representation that allows for the discontinuities generated by fixed adjustment costs. Third, I use this NN-KS method to extend household preferences in the model of Bloom et al. (2018) and show that a pure uncertainty shock can generate comovement in output, consumption, investment, and employment on impact, without an accompanying adverse first-moment shock.

Many economic models do not yield explicit formulae for agents' decisions, prices, and allocations. This difficulty is particularly acute for heterogeneous firm models because closed-form solutions for market-clearing prices cannot be derived. This is problematic for the Krusell-Smith algorithm because it relies on a long simulation of the stochastic economy wherein, in each period, we must find the market-clearing equilibrium prices (such as wages) numerically. This contrasts with standard incomplete market, household-centric heterogeneous agent models, where price determination is often simpler even under aggregate uncertainty.

In more detail, the standard approach requires solving the firm's decision problem given a guessed price in each simulation period. We then use the solved policy rules to obtain the distribution of firms and consequent aggregate quantities to check if market clearing is satisfied. Since the initial guess is rarely correct, we must iteratively repeat this process to find the market-clearing prices before moving to the next period. We then carry the resulting distribution of firms forward and restart the numerical search for the next period's prices, repeating this over the entire stochastic simulation of 5,000 to 10,000 periods. This process is incredibly demanding; for

[1] Terry (2017) systematically compares five solution methods for heterogeneous firm models—the traditional Krusell-Smith algorithm (Krusell and Smith (1998), Krusell and Smith (1997)), Parameterized Distributions (Algan et al., 2008), Explicit Aggregation (Den Haan and Rendahl, 2010), Projection plus Perturbation (Reiter, 2009), and Parameterization plus Perturbation (Winberry, 2018).

instance, solving the model in Bloom et al. (2018) can take four days to complete.

The novelty of the proposed method is that it resolves this exact bottleneck by rendering the repetitive solution of the firm's decision problem unnecessary during the price search. Specifically, I include prices directly as arguments in the firm's policy functions alongside moments of distribution and precompute these policy functions before simulation. This allows the simulation loop to determine the market-clearing price quickly via the policy function, bypassing the time-consuming optimization problem required to derive decision rules at every guess. The simulation of Bloom et al. (2018) model, which takes 290 minutes for 5,000 periods under the standard Krusell-Smith method, can be simulated in just 1.5 minutes.

However, precomputing policies on a grid becomes costly as the number of individual states, aggregate moments, and prices increases. The number of entries in the policy table grows multiplicatively with these dimensions. I therefore approximate both the value function and the price-conditioned policy function with neural networks. Once trained, evaluating the policy for any candidate price becomes a simple forward pass through the network. A key contribution is a neural-network architecture that represents discontinuous policy rules, such as (S, s) rules. Specifically, I decompose the firm's decision into a discrete choice of whether to adjust and a continuous choice of how much to adjust, conditional on adjustment. The discrete policy network selects an adjustment regime, and the conditional policy network determines the action within that regime. This representation preserves exact inaction and allows the policy to jump at adjustment thresholds. NN–KS completes the Bloom et al. (2018) application in 183 minutes. Its principal advantage over the accelerated grid method is the ability to accommodate additional state variables and equilibrium prices.

The economic application uses this computational gain to revisit aggregate comovement after an uncertainty shock. In the baseline Bloom et al. (2018) model, output, investment, and employment fall, while consumption rises on impact. The authors discuss alternative household preferences, such as the GHH preference (Greenwood et al. (1988)), as a possible way to address this consumption response, but leave this extension unexplored because it requires an additional state variable and equilibrium price, which significantly increases the computational burden. The NN–KS makes it practical to examine this extension.

The results show that introducing a finite Frisch elasticity into the original specification, which assumes an infinite Frisch elasticity, is sufficient to generate comovement. A finite Frisch elasticity allows the real wage to fall with employment, cushioning the contraction in labor. The lower wage makes continued hiring more attractive, so the smaller decline in employment cushions the fall in the marginal product of capital and supports investment. Through goods-market clearing, the smaller reduction in investment spending allows consumption to fall alongside output.

In addition to this wage channel, complementarity between consumption and hours worked further moderates the decline in investment. Under GHH preferences, where consumption–hours complementarity is strong, the decline in hours lowers the marginal utility of consumption,

more than offsetting the increase caused by lower consumption. The resulting decline in current marginal utility reduces the utility cost of current investment relative to future returns for firms near their adjustment thresholds, causing aggregate investment to rise on impact. The quasi-separable specification of Bilbiie et al. (2025) allows for an intermediate degree of consumption–hours complementarity. In this application, it generates a larger decline in consumption than separable preferences with finite Frisch elasticity while preserving the decline in investment.

Related Literature. The paper builds on the literature on solution methods for heterogeneous-agent models with aggregate uncertainty. Den Haan (2010b) compares solutions of the original KS model, and Terry (2017) examines heterogeneous-firm models. Alternatives include parameterized distributions (Algan et al., 2008), explicit aggregation (Den Haan and Rendahl, 2010), and methods combining distributional approximations with perturbation (Reiter, 2009; Winberry, 2018). More recent approaches also target the cost of finding equilibrium prices (Bakota, 2023; Lee, 2026). They propose alternative ways to mitigate this bottleneck by advancing the simulation without imposing market clearing in every period. My approach retains the standard KS simulation and its period-by-period market-clearing search, while removing repeated firm-level optimization from that search. **?** adopt a similar strategy of treating prices as state variables, though they compute a Sequential Restricted Perceptions Equilibrium without the distribution included as a state variable. In contrast, my method retains the standard KS approximation to a rational-expectations equilibrium, using distributional moments in the forecasting rules and imposing market clearing in each simulation period.

The neural network extension is related to the growing literature on deep learning for dynamic economic models. My approach uses neural networks to approximate the price-conditioned policy function and value function, mitigating the curse of dimensionality while otherwise retaining the standard KS structure. However, enabling neural networks to handle (S, s)-type discontinuous policy functions is one of my contributions to this literature. This builds on the use of classification for discrete decisions in Maliar and Maliar (2022). Combining a discrete policy with conditional continuous policies accommodates discontinuous adjustment decisions while preserving exact inaction.

The application contributes to the literature on uncertainty shocks and aggregate comovement. Bloom et al. (2018) study how uncertainty interacts with capital and labor adjustment costs. Basu and Bundick (2017) obtain comovement through nominal price rigidity and countercyclical markups in a model of effective demand. The application here shows that a pure uncertainty shock can also generate simultaneous impact declines in output, consumption, investment, and employment under flexible prices, through household preferences and the general-equilibrium response to firms' adjustment decisions.

Jaimovich and Rebelo (2009) show how preferences, investment adjustment costs, and variable capital utilization can generate comovement after news about future fundamentals. Their

shocks concern first moments; the present application examines second-moment shocks. GHH and quasi-separable preferences provide ways to vary consumption–hours complementarity and wealth effects in this exercise (Greenwood et al., 1988; Bilbiie et al., 2025).

Organization of the paper. Section 2 presents the model of Bloom et al. (2018) and explains the KS method and its simulation bottleneck. Section 3 develops policy precomputation, reports its computational gains, and explains the role of price errors in preserving accuracy. Section 4 presents NN–KS for the same model and evaluates its performance and accuracy. Section 5 studies household preferences and uncertainty-shock dynamics in the Bloom et al. (2018) model. Section 6 concludes. The Supplemental Appendix contains model specifications, implementation details, and additional results.

2 The Model and the Krusell–Smith Method

This section presents the model of Bloom et al. (2018) and explains how it is solved using the Krusell–Smith (KS) method. The model combines aggregate and idiosyncratic uncertainty with capital and labor adjustment costs. Its equilibrium prices must be determined jointly with firms' decisions, making the simulation stage of the KS method particularly costly.

2.1 Model

Time is discrete. A unit mass of heterogeneous firms produces a homogeneous good using a Cobb–Douglas technology in capital and labor:

$$y = z\varepsilon k^{\alpha} n^{\nu}. \tag{1}$$

Here, z is aggregate productivity, ε is idiosyncratic productivity, k is capital, and n is current employment.

Productivity processes. Both aggregate and idiosyncratic productivity follow persistent stochastic processes with time-varying volatility:

$$\log z' = \rho_z \log z + \sigma_z u'_z, \qquad u'_z \sim N(0,1), \tag{2}$$

$$\log \varepsilon' = \rho_\varepsilon \log \varepsilon + \sigma_\varepsilon u'_\varepsilon, \qquad u'_\varepsilon \sim N(0,1). \tag{3}$$

The time-varying volatilities σ_z and σ_ε capture switches between low-uncertainty and high-uncertainty regimes governed by a Markov process. These uncertainty states become part of the aggregate state space. I use Z to collect the exogenous aggregate states. In the numerical

implementation, a common uncertainty regime governs both productivity processes, and the lagged uncertainty regime is also included, so that $Z = (z, \sigma, \sigma_{-1})$.

Capital dynamics and adjustment costs. Capital evolves according to the standard law of motion $k' = (1 - \delta_k)k + i$. Investment is subject to adjustment costs AC^k, which include a fixed cost proportional to output and a term capturing partial irreversibility:

$$\mathrm{AC}^k = \mathbf{1}_{|i|>0}\, y\, F^K + S|i|\, \mathbf{1}_{i<0}. \tag{4}$$

Here, $\mathbf{1}_{(.)}$ is an indicator function, F^K is the fixed disruption cost parameter, and S is the resale loss fraction for disinvested capital.

Labor dynamics and adjustment costs. Labor also faces adjustment frictions and evolves as

$$n = (1 - \delta_n)n_{-1} + s, \tag{5}$$

where n_{-1} is the previous period's employment, δ_n is an exogenous separation rate, and s represents net hiring or firing. Labor adjustment costs AC^n include a fixed cost proportional to output and a variable cost related to the hiring or firing flow:

$$\mathrm{AC}^n = \mathbf{1}_{|s|>0}\, y\, F^L + |s|\, H\, w. \tag{6}$$

The parameter F^L is the fixed disruption cost for labor adjustment. The second term is a linear hiring or firing cost expressed as a fraction of the aggregate wage w. Both capital and labor adjustment costs absorb goods. Because labor adjustment depends on previous employment, n_{-1} is an additional endogenous state variable for the firm.

Households and equilibrium prices. A representative household owns the firms, supplies labor, and has preferences

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \big(\log C_t - \chi N_t\big), \qquad 0 < \beta < 1, \quad \chi > 0. \tag{7}$$

Consumption is the numeraire. Let p denote the Lagrange multiplier on the household's budget constraint, which equals the marginal utility of current consumption. The household's first-order conditions imply

$$p = \frac{1}{C}, \qquad w = \frac{\chi}{p}, \qquad M_{t,t+1} = \beta \frac{p_{t+1}}{p_t}, \tag{8}$$

where $M_{t,t+1}$ is the stochastic discount factor used to value firm dividends. I refer to p as the price of the consumption good in marginal-utility units. Only one of p and w must be determined numerically in the baseline model; the other follows from (8). Section 5 extends household preferences and allows both prices to be determined independently.

Firm problem and state space. Let μ be the cross-sectional distribution of firms over their individual states (k, n_{-1}, ε) after current shocks are observed and before current choices are made. Multiplying the firm's value in goods units, V, by marginal utility gives $\tilde{V} \equiv pV$. The Bellman equation in marginal-utility units is

$$\begin{aligned}\tilde{V}(k, n_{-1}, \varepsilon; Z, \mu) = \max_{i,n} \Big\{ & p(Z,\mu)\left[y - w(Z,\mu)n - i - \mathrm{AC}^k - \mathrm{AC}^n\right] \\ & + \beta \mathbb{E}\left[\tilde{V}(k', n, \varepsilon'; Z', \mu')\right] \Big\}.\end{aligned} \tag{9}$$

The choices satisfy the capital and labor laws of motion. Each firm is atomistic and takes aggregate prices and the evolution of μ as given. The individual state is (k, n_{-1}, ε), and the aggregate state consists of Z and μ. The equilibrium price mapping and distributional law of motion are

$$\mu' = \Gamma_\mu(Z, \mu), \qquad p = \Gamma_p(Z, \mu), \tag{10}$$

with the distributional transition incorporating firm decisions and realized productivity shocks.

For a given firm policy, aggregate output, investment, employment, and adjustment costs are

$$\begin{aligned} Y &= \int z\varepsilon k^\alpha n^\nu \, d\mu, & I &= \int i \, d\mu, \\ N &= \int n \, d\mu, & AC &= \int \left(\mathrm{AC}^k + \mathrm{AC}^n\right) d\mu. \end{aligned} \tag{11}$$

Goods-market clearing requires

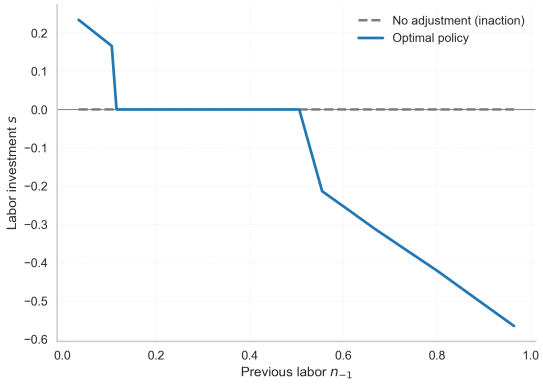
$$C = Y - I - AC. \tag{12}$$

A recursive competitive equilibrium consists of value and policy functions, a price mapping, and a law of motion for the distribution such that firms solve (9), the household optimizes, and markets clear. Firms choose their optimal actions conditional on p. These actions determine aggregate consumption $C(p)$ through (12), while household optimality requires $p = 1/C(p)$. Thus, in each simulation period, one must find the value of p that solves this implicit equation.

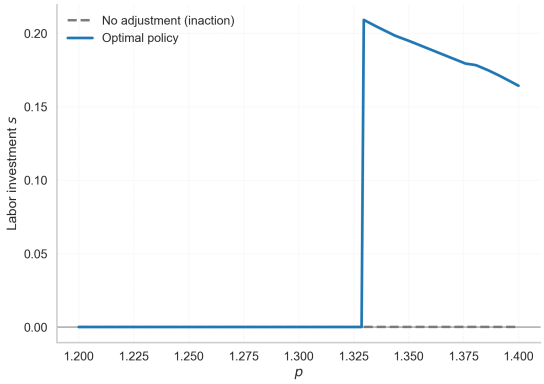
Non-smooth adjustment costs. The model includes fixed costs for adjusting capital or labor, a penalty for selling capital, and linear hiring or firing costs. These features create kinks and sudden jumps in the firm's optimal policy. A firm pays a fixed cost only if the benefit from adjustment is large enough. Otherwise, it chooses *inaction*, allowing capital to depreciate and employment to shrink through natural separations. This leads to an (S, s) adjustment rule. Following Bloom et al. (2018), I set the capital fixed-cost parameter F^K in equation (4) to 0.0, so there is no fixed cost of capital adjustment.

Figure 1 illustrates the (S, s) pattern of employment adjustment as a function of previous employment and the candidate price. A region of inaction separates the adjustment regimes, with jumps at their boundaries. The numerical method must account for this discrete adjustment deci-

Figure 1: Evidence of an S–s type adjustment rule in labor investment



(a) Labor adjustment as a function of previous labor n_{t-1}



(b) Labor adjustment as a function of price p

sion.

2.2 The Krusell–Smith Method

Following Krusell and Smith (1998), firms approximate the infinite-dimensional distribution μ with a finite set of moments. In the baseline model, the moment is aggregate capital,

$$K = \int k \, d\mu. \tag{13}$$

Let θ collect the parameters of the forecasting rules. Firms use

$$K' = \Gamma_K(Z, K; \theta), \tag{14}$$

$$\widehat{p} = \Gamma_p(Z, K; \theta), \qquad \widehat{w} = \chi / \widehat{p}. \tag{15}$$

For example, these rules can be specified as $\log K' = a_K(Z) + b_K(Z) \log K$ and $\log \widehat{p} = a_p(Z) + b_p(Z) \log K$. The forecasting rules approximate the distributional state when evaluating continuation values. Simulation retains the cross-sectional distribution and determines current market-clearing prices in every period.

Step 1: Solve the firm's problem under the forecasting rules. For fixed θ, solve

$$\begin{aligned} \tilde{V}_\theta(k, n_{-1}, \varepsilon; Z, K) = \max_{i,n} \Big\{ & \widehat{p}\big[y - \widehat{w}n - i - \mathrm{AC}^k - \mathrm{AC}^n(\widehat{w})\big] \\ & + \beta \mathbb{E}\big[\tilde{V}_\theta(k', n, \varepsilon'; Z', \Gamma_K(Z, K; \theta))\big] \Big\}. \end{aligned} \tag{16}$$

The value function is computed on a grid, using interpolation only for the aggregate capital. The current price is supplied by Γ_p, and future aggregate capital is supplied by Γ_K throughout this

step.

The numerical implementation uses 91 points for capital, 37 for previous employment, 5 for idiosyncratic productivity, 5 for aggregate productivity, 2 for the current uncertainty regime, 2 for the lagged uncertainty regime, and 10 for aggregate capital:

$$91 \times 37 \times 5 \times 5 \times 2 \times 2 \times 10 = 3{,}367{,}000 \quad \text{state points.}$$

The capital and labor grids are constructed so that adjacent grid points correspond to depreciated values. For example, $k_1 = (1 - \delta_k)k_2$ for neighboring capital points, so moving from k_2 to k_1 represents exact inaction without paying the fixed cost.

Step 2: Simulate the economy and clear markets. In period t, the simulator knows (Z_t, μ_t) and computes $K_t = \int k \, d\mu_t$. Before updating the distribution, we need to find the market clearing price p. Forecasting rule for p is not necessarily consistent with a true market clearing price in the middle of solving the model. We find the price by iterating the process, in which guessing a candidate price, computing firms' optimal policy, aggregating the policy and check if the household optimal condition is satisfied.

For compact notation, let $x_f = (k, n_{-1}, \varepsilon)$ denote the firm's individual state, let $k_0 = (1 - \delta_k)k$, and let $a = (k', n)$ denote its action. Investment is then $i = k' - k_0$. At a candidate current price p, set $w = \chi/p$ and define

$$\begin{aligned} R_\theta(x_f, a; Z, K, p) = p\big[y - (\chi/p)n - (k' - k_0) - \mathrm{AC}^k - \mathrm{AC}^n(\chi/p)\big] \\ + \beta \mathbb{E}\big[\tilde{V}_\theta(k', n, \varepsilon'; Z', \Gamma_K(Z, K; \theta))\big]. \end{aligned} \tag{17}$$

The associated policy is

$$g_\theta(x_f; Z, K, p) \in \underset{a \in \mathcal{A}(x_f, Z)}{\arg\max} \, R_\theta(x_f, a; Z, K, p), \tag{18}$$

where $\mathcal{A}(x_f, Z)$ is the feasible set of capital and employment choices. The value function and forecasting rules remain fixed during the price search. [2]

Using $g_\theta(x_f; Z, K, p)$ and the actual distribution μ, calculate aggregate quantities from (11) and implied consumption $C(p) = Y(p) - I(p) - AC(p)$. The current price must satisfy

$$p = \frac{1}{C(p)}. \tag{19}$$

[3] Once the price search is complete, the simulator updates the distribution using firms' decisions

[2] At a given action, the continuation term in (17) is unchanged by the candidate current price. The optimal action generally changes with that price. Each candidate therefore requires a current decision problem, rather than a new value-function iteration.

[3] Because fixed adjustment costs can generate jumps in excess demand, the original implementation evaluates excess

and realized shocks.

Step 3: Update the forecasting rules. After simulating the economy and discarding an initial transient, update θ using realized prices and aggregate capital. Repeat the solution and simulation steps until the forecasting rules converge, and evaluate their predictive accuracy using the simulated economy.

The computational bottleneck. The main simulation cost arises in Step 2. For every candidate price, the standard method solves (18) for the individual states used in the simulation, aggregates the resulting decisions, and checks market clearing. A new price candidate requires another maximization for each individual state. This process starts again in every simulation period. In the Bloom et al. (2018) model, a 5,000-period simulation takes 290 minutes using a Fortran implementation with OpenMP parallelization on an AMD Ryzen 7 7700 processor. The next section removes this repeated firm-level optimization from the price search.

3 Accelerating the Krusell–Smith Method with Precomputation

The proposed method computes firms' responses to candidate prices before simulation. It retains the KS iteration on forecasting rules and the period-by-period search for market-clearing prices. Its computational gain comes from replacing the repeated maximization inside that search with evaluation of a precomputed policy.

3.1 Precomputing the Price-Conditioned Policy

After solving for $\tilde{V}_\theta$, construct a grid over individual states, aggregate states, aggregate capital, and candidate prices. At each grid point, solve

$$\bar{g}_\theta(x_f; Z, K, p) \in \underset{a \in \mathcal{A}(x_f, Z)}{\arg\max}\, R_\theta(x_f, a; Z, K, p). \tag{20}$$

The continuation value and forecasting rules are the same as in the KS method. The difference is when the conditional optimization is performed: optimal decisions are computed and stored before simulation begins. The policy table is then evaluated at each candidate price, using interpolation between grid points where needed.

During simulation, a trial price requires only evaluating the stored policy at (x_f, Z_t, K_t, p) for each relevant individual state. Aggregate these decisions using the current distribution μ_t, evaluate the market-clearing residual, and update the trial price. Once the price search is complete, use the policy at that price to advance the distribution. The same precomputed table is used across

demand on a grid of candidate prices and finds a zero of its interpolated excess demand.

Table 1: Computation time in the Bloom et al. (2018) model

Model	Method	Total	VFI	Precompute policy	Simulation
Bloom et al. (2018)	KS	6,141 min ($\approx$ 4 days)	7 min	–	290 min
Bloom et al. (2018)	Proposed	108 min	7 min	25 min	1.5 min

price candidates and simulation periods. It is updated when the outer iteration changes the forecasting rules.

The current price is an additional argument of the policy function because firms must respond to every price considered in the market-clearing search. A policy computed only at the forecast price $\Gamma_p(Z, K; \theta)$ would not provide this response. Including the candidate price directly makes it possible to move the conditional optimization out of the simulation loop.

3.2 Computation Time

Table 1 reports computation time for the baseline KS method and the proposed method. Total computation time falls from 6,141 minutes, approximately four days, to 108 minutes. The main source of the speedup is the simulation step: a 5,000-period simulation decreases from 290 minutes to 1.5 minutes because the proposed method avoids repeatedly solving the firm's maximization problem during the market-clearing price search. The value-function iteration takes 7 minutes in both methods, so the improvement does not come from solving the Bellman equation faster.

In this application, I use 25 grid points for p. Combining 3,367,000 state points with 25 price-grid points requires solving the firm problem (20) at 84,175,000 state–price combinations. Fixed adjustment costs further require a brute-force comparison across all discrete capital- and labor-adjustment regimes at each combination. These features account for the 25-minute precomputation time in the first outer iteration.

Subsequent iterations use the policy from the previous outer loop as a warm start, since small updates to the forecasting rules leave the previous policy close to optimal. This substantially reduces the cost of subsequent precomputation steps, bringing total computation time to 108 minutes. Thus, the upfront cost of constructing the policy table is more than offset by the reduction in simulation time.

3.3 Price Errors and Accuracy

Replacing repeated maximization with policy evaluation is the source of the simulation speedup. The accuracy of this approach depends on whether the precomputed policy captures firms' responses to prices. The placement of price-grid points is therefore crucial.[4]

[4]This issue is especially important in models in which optimal actions are sensitive to equilibrium prices, as discussed in Koby and Wolf (2020) and Winberry (2021).

A price grid centered on the forecasting rule. A fixed grid of price levels must cover the variation induced by many aggregate states. For example, in the Bloom et al. (2018) implementation, the equilibrium price ranges from 1.1 to 1.7, requiring a uniform grid that spans the entire interval. However, when the forecasting rule is informative, equilibrium prices tend to remain close to the corresponding forecast. A grid over the full range of price levels can therefore allocate many points far from the relevant price at a given aggregate state while providing relatively coarse local resolution.

To concentrate grid points around the forecast, define the deviation of the candidate price from its forecast as

$$p_{\text{error}} = p - \Gamma_p(Z, K; \theta). \tag{21}$$

The price and wage corresponding to a deviation-grid point are

$$p = \Gamma_p(Z, K; \theta) + p_{\text{error}}, \qquad w = \frac{\chi}{\Gamma_p(Z, K; \theta) + p_{\text{error}}}. \tag{22}$$

The wage is recomputed consistently for every candidate p. During the search, p_{error} is a candidate-price deviation; it becomes a realized forecast error only at the market-clearing price.

Figure 2 illustrates this construction. The lower axis places grid points around zero in the deviation p_{error}. The dotted lines map these points to candidate price levels on the upper axis according to $p = \Gamma_p(Z, K; \theta) + p_{\text{error}}$. The larger blue point marks the forecast itself. As the aggregate state changes, the price grid moves with the forecast, while the grid of deviations remains fixed.

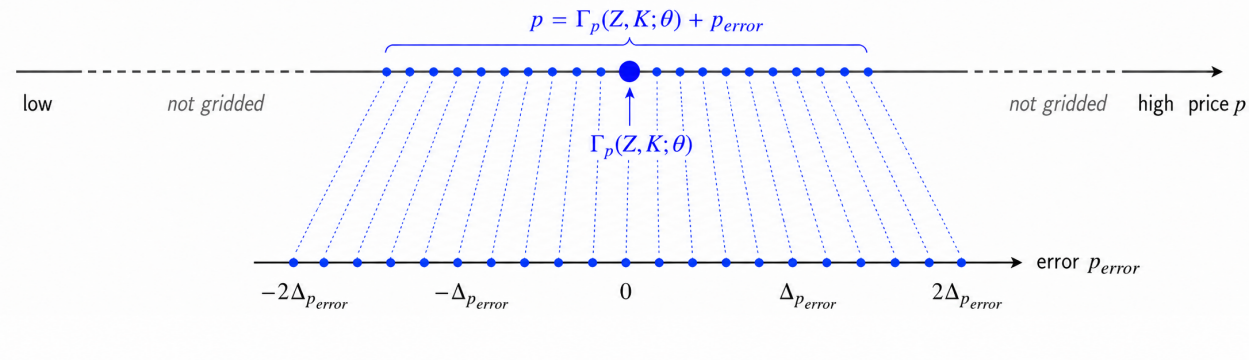


Figure 2: Price grid centered on the forecast price

The portions of the upper axis without grid points show that the table covers a local range of price levels at each aggregate state. This concentration is important because the policy is precomputed only at discrete price nodes, whereas the market-clearing search generally evaluates candidate prices between those nodes. The accuracy of interpolation in the price dimension affects the aggregate quantities used to determine the market-clearing price. With a given number of nodes, the forecast-error grid provides finer price resolution around the forecast and can re-

duce local interpolation error relative to a grid spanning the full range of price levels. The interval must still cover the candidates required by the search, including those encountered before the forecasting rules converge, while respecting positive prices and feasible consumption.

Using these coordinates, the stored policy is represented as

$$\widetilde{g}_\theta(x_f; Z, K, p_{\text{error}}) = \bar{g}_\theta\big(x_f; Z, K, \Gamma_p(Z, K; \theta) + p_{\text{error}}\big). \tag{23}$$

Precomputation solves the same conditional optimization at the price implied by each deviation-grid point. During simulation, each trial price is converted to its deviation from the current forecast, and the policy is evaluated between the surrounding deviation-grid nodes. Precomputation eliminates repeated optimization, while the forecast-error grid helps preserve the accuracy of the policy evaluation needed during market clearing.

Implementation in the Bloom et al. (2018) price search. In the Bloom et al. (2018) model, fixed adjustment costs generate (S, s) policies, so excess demand can jump when a small price change induces a positive mass of firms to switch adjustment regimes. Their implementation therefore evaluates excess demand on a grid of candidate prices, constructs a continuous interpolated excess-demand schedule from these grid values, and searches for the price at which the interpolated schedule is zero.

The error-price formulation is useful in this procedure because it concentrates the price nodes around the forecast, where the market-clearing price is likely to lie. For a fixed number of nodes, this concentration narrows the price interval over which excess demand is interpolated near the solution and thereby improves the accuracy of the market-clearing price. Appendix A.2 separately examines the importance of the error-price formulation in the Khan–Thomas model, where the price search does not interpolate excess demand.

3.4 Limitations of Precomputation

The gain during simulation comes at an upfront cost. If N_{state} denotes the number of state-grid combinations, N_e the number of price-deviation combinations, and d_a the number of stored policy outputs, precomputation requires approximately $N_{\text{state}} N_e$ conditional optimizations, and storage is proportional to $d_a N_{\text{state}} N_e$. These costs must be included when evaluating total runtime.

In the present model, the policy-function matrix requires roughly 1.25 GB of memory. Additional individual state variables, aggregate moments, or equilibrium prices multiply these storage requirements and increase the cost of precomputation. For example, the preference extensions in Section 5 require an additional equilibrium price, the wage w, and an additional distributional moment, lagged aggregate employment N_{-1}. Relative to the baseline specification, these additions introduce two further dimensions into the policy table. These computational limitations motivate the neural-network approach developed in Section 4.

4 The NN–KS Method

In Section 3, the price-conditioned policy $\widetilde{g}_\theta(x_f; Z, K, p_{\text{error}})$ in (23) is stored as a table on a state–price grid. NN–KS approximates the same policy using neural networks—specifically, using a discrete policy network g_d and a conditional policy network g_c. The economic problem, forecasting rules, market-clearing equation, distribution update, and outer KS iteration remain unchanged. The acceleration principle is also the same: firms' responses to candidate prices are learned before simulation, so the market-clearing search evaluates policies without repeatedly solving firms' optimization problems. Neural networks reduce the storage and interpolation costs of a large policy table, which contains 84,175,000 state–price combinations and occupies approximately 1.25 GB in the grid-based implementation.

Throughout this section, $x_f = (k, n_{-1}, \varepsilon)$ and $Z = (z, \sigma, \sigma_{-1})$ retain their definitions from Section 2, and K is the aggregate capital. The forecasting rules are $\Gamma_K(Z, K; \theta)$ and $\Gamma_p(Z, K; \theta)$. The policy networks take the price deviation p_{error} as an input; the candidate price and corresponding wage are recovered as

$$p = \Gamma_p(Z, K; \theta) + p_{\text{error}}, \qquad w = \chi / p,$$

as in (22). A value network approximates the Step 1 value function $\tilde{V}_\theta(x_f; Z, K)$ in (16). Its parameters ω_V, and the policy-network parameters ω_d and ω_c, are distinct from the forecasting-rule parameters θ. For each fixed θ, policy training covers the candidate-price range, whereas value updates use the forecast price, $p_{\text{error}} = 0$.

4.1 Decomposing the Single max: Discrete and Conditional Policies

Following and extending the classification approach of Maliar and Maliar (2022), I separate the choice of an adjustment regime from the conditional continuous choices of capital and employment. The discrete policy network g_d selects whether to adjust. Given this regime, the conditional policy network g_c determines (k', n) and imposes exact inaction in any dimension that is not adjusted.

Why separate discrete and continuous choices? An (S, s) policy is discontinuous: the optimal choice jumps when the firm crosses an adjustment threshold. A standard feedforward neural network with continuous activation functions produces a continuous mapping and therefore cannot represent such a policy exactly with a single output network.[5]

A single continuous policy network also cannot reliably represent inaction. Exact inaction requires the network to output exactly $k' = (1 - \delta_k)k$ for capital and $n = (1 - \delta_n)n_{-1}$ for labor. Even a small approximation error produces nonzero investment or labor adjustment; on the labor side,

[5] Classical uniform-approximation results apply to continuous target functions; see Cybenko (1989) and Hornik (1991).

it can therefore incorrectly trigger the fixed adjustment cost. Treating sufficiently small adjustments as zero would instead require an arbitrary numerical threshold. I avoid both problems by using a discrete network to choose the adjustment regime and a conditional continuous network to choose the action within that regime. The resulting composite policy can jump at adjustment boundaries and imposes exact inaction by construction.

Four adjustment regimes. Define $k_0 = (1-\delta_k)k$ and $n_0 = (1-\delta_n)n_{-1}$, and set

$$p = \Gamma_p(Z, K; \theta) + p_{\text{error}}.$$

Write the candidate return in (17) directly as a function of (k', n) for notational simplicity:

$$\begin{aligned} R_\theta(k', n) \equiv p\big[y - (\chi/p)n - (k' - k_0) - \mathrm{AC}^k - \mathrm{AC}^n(\chi/p)\big] \\ + \beta \mathbb{E}\big[\tilde{V}_\theta(k', n, \varepsilon'; Z', \Gamma_K(Z, K; \theta))\big]. \end{aligned} \tag{24}$$

Here and below, the common arguments $(x_f; Z, K, p_{\text{error}})$ are suppressed.

For numerical implementation, I decompose (24) into four regimes:

$$\max_{k', n} R_\theta(k', n) = \max\big\{R_\theta^{00}, R_\theta^{10}, R_\theta^{01}, R_\theta^{11}\big\} = \max_{r \in \mathcal{R}} R_\theta^r. \tag{25}$$

Here, $r = (r_k, r_n)$ belongs to $\mathcal{R} \equiv \{(0,0), (1,0), (0,1), (1,1)\}$. The first component r_k indicates capital adjustment, and the second component r_n indicates labor adjustment. In either component, 1 denotes adjustment and 0 denotes inaction; the superscripts 00, 10, 01, and 11 abbreviate these pairs.

For each $r \in \mathcal{R}$, R_θ^r denotes the maximum of the return in (24) subject to the restrictions associated with regime r. Within each branch, k' and n are selected as follows:[6]

$$\begin{aligned} r = (0,0): &\quad k' = k_0, &&\quad n = n_0, && \text{(no capital or labor adjustment)}, \\ r = (1,0): &\quad k' \text{ is chosen continuously}, &&\quad n = n_0, && \text{(capital adjustment only)}, \\ r = (0,1): &\quad k' = k_0, &&\quad n \text{ is chosen continuously}, && \text{(labor adjustment only)}, \\ r = (1,1): &\quad k' \text{ and } n \text{ are chosen continuously}, && && \text{(capital and labor adjustment)}. \end{aligned} \tag{26}$$

Hence, the firm's single maximization separates into the outer discrete choice of the adjustment regime in (25), approximated by g_d, and the regime-specific choices in (26), approximated by g_c.

[6] Mathematically, regime $(1,1)$ nests the other regimes because k' and n may equal k_0 and n_0, respectively. Under the continuous neural-network approximation, however, such exact equalities do not arise in practice.

Discrete policy. The discrete policy network outputs four softmax scores:

$$\left(\pi_r(x_f; Z, K, p_{\text{error}})\right)_{r\in\mathcal{R}} = g_d(x_f; Z, K, p_{\text{error}}; \omega_d), \qquad \sum_{r\in\mathcal{R}} \pi_r = 1.$$

These normalized scores represent the network's predicted probabilities for the four adjustment regimes. The actual regime is selected deterministically by

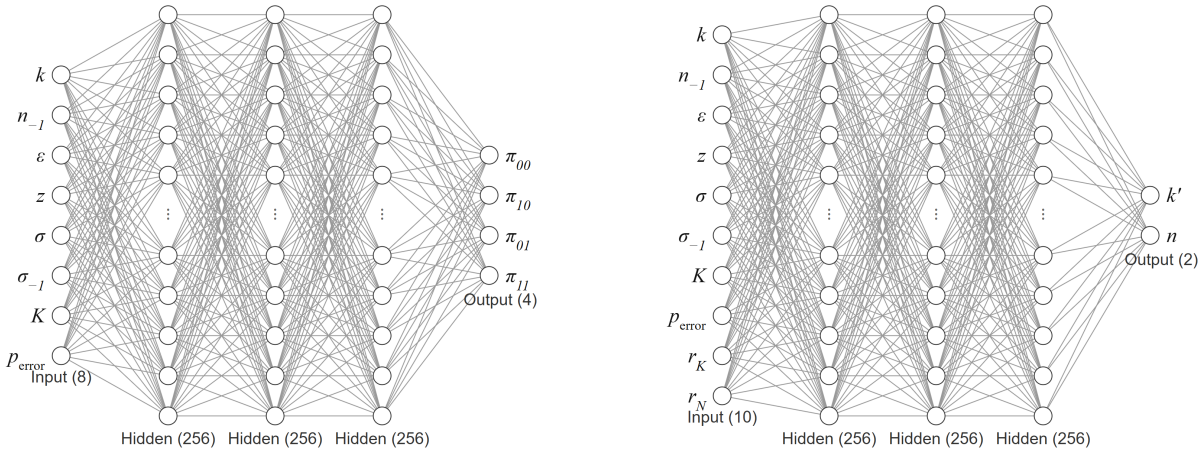
$$\widehat{r}(x_f; Z, K, p_{\text{error}}) = \arg\max_{r\in\mathcal{R}} \pi_r(x_f; Z, K, p_{\text{error}}).$$

Conditional continuous policy. The conditional policy network receives the same state and price deviation, together with the selected regime encoded by its two binary digits:

$$(k', n) = g_c(x_f; Z, K, p_{\text{error}}, r; \omega_c).$$

The regime restrictions impose $k' = k_0$ in regimes 00 and 01. Likewise, they impose $n = n_0$ in regimes 00 and 10 .

Figure 3: Neural network architectures for discrete and conditional policies.



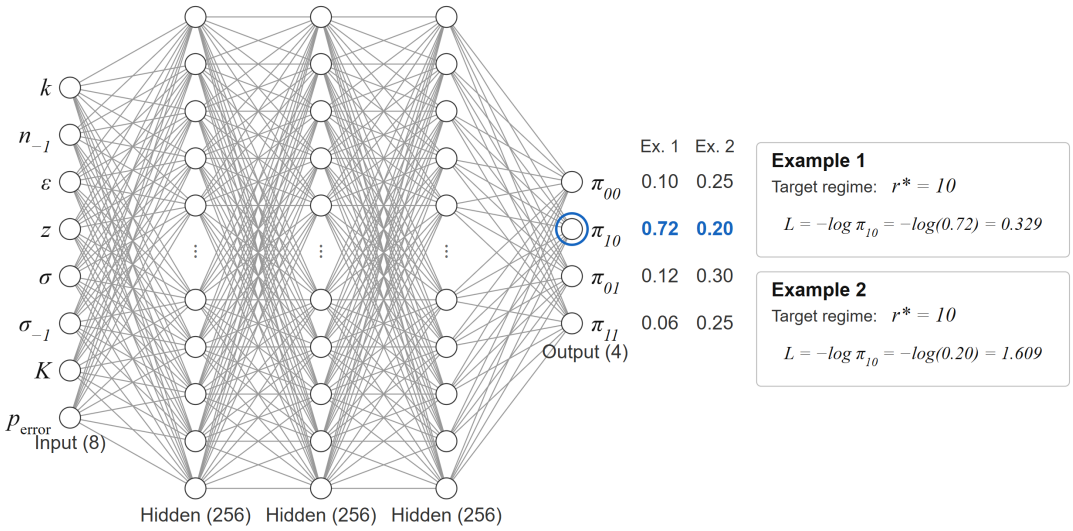
(a) Discrete policy network g_d

(b) Conditional policy network g_c

Figure 3 shows the two networks. Their common inputs are $(k, n_{-1}, \varepsilon, z, \sigma, \sigma_{-1}, K, p_{\text{error}})$. The discrete network selects a regime, and the conditional network supplies capital and employment choices within it. Simulation requires only these forward evaluations at each candidate price.

Training the discrete policy. For fixed θ, training the discrete policy network g_d is a standard supervised classification problem following Maliar and Maliar (2022). Given the economic state and a candidate price, the network learns to assign the firm to one of the four regimes $r \in \{00, 10, 01, 11\}$. First, I construct the training label by comparing the regime-specific candidate

Figure 4: Training the discrete policy network



returns implied by the current value and conditional-policy approximations. For each training observation, I evaluate the return from all four regimes with current value function and conditional continuous policy function, and select the regime that delivers the highest value:

$$r^*(x_f; Z, K, p_{\text{error}}) \in \arg\max_{r \in \mathcal{R}} R^r_\theta(x_f; Z, K, p_{\text{error}}).$$

Here, R^r_θ is the sum of the current return and the continuation value when the firm follows regime r. The maximizing regime becomes the target label r^*. [7]

The discrete network outputs four probabilities, $\{\pi_{00}, \pi_{10}, \pi_{01}, \pi_{11}\}$, and is trained to place a high probability on the target regime. For a batch of B observations indexed by j, it minimizes the cross-entropy loss

$$\mathcal{L}_{\text{disc}}(\omega_d) = -\frac{1}{B}\sum_{j=1}^{B} \log \pi_{r^*_j}(x_{f,j}; Z_j, K_j, p_{\text{error},j}).$$

Figure 4 illustrates this training mechanism. The network takes $(k, n_{-1}, \varepsilon, z, \sigma, \sigma_{-1}, K, p_{\text{error}})$ as inputs and returns a probability for each of the four regimes. The figure shows two examples in which the target regime is $r^* = 10$. When the network assigns probability $\pi_{10} = 0.72$ to that regime, the loss is only $-\log(0.72) \approx 0.329$. When it assigns probability $\pi_{10} = 0.20$, the loss rises to $-\log(0.20) \approx 1.609$. The cross-entropy loss therefore penalizes low probability on the value-maximizing regime and trains g_d to imitate the firm's optimal discrete decision.

[7] Because the conditional policy and value approximations change during training, I recompute these labels as the two networks improve.

Training the conditional policy. For each sampled state and price deviation, let $\widehat{r}_j$ be the regime selected by g_d, and set

$$(k'_j, n_j) = g_c(x_{f,j}; Z_j, K_j, p_{\text{error},j}, \widehat{r}_j; \omega_c), \qquad i_j = k'_j - (1-\delta_k)k_j.$$

Holding the value network and discrete choices fixed, update ω_c to maximize the return at these choices, or equivalently minimize

$$\mathcal{L}_{\text{cont}}(\omega_c) = -\frac{1}{B}\sum_{j=1}^{B} R_\theta\left(x_{f,j}, (k'_j, n_j); Z_j, K_j, \Gamma_p(Z_j, K_j; \theta) + p_{\text{error},j}\right).$$

The continuation value uses the value network. Training focuses on the predicted regimes.

Value network and training schedule. The value network satisfies

$$\tilde{V}_{\text{nn}}(x_f; Z, K; \omega_V) \approx \tilde{V}_\theta(x_f; Z, K).$$

The target, which is the right-hand side of the Bellman equation, uses the forecast price, as in Step 1 of Section 2.2. For each sampled state, set

$$\widehat{p}_j = \Gamma_p(Z_j, K_j; \theta), \qquad \widehat{w}_j = \chi/\widehat{p}_j, \qquad p_{\text{error},j} = 0,$$

and evaluate the composite policy at this zero deviation:

$$\widehat{r}_j = \arg\max_{r \in \mathcal{R}} \pi_r(x_{f,j}; Z_j, K_j, 0), \qquad (k'_j, n_j) = g_c(x_{f,j}; Z_j, K_j, 0, \widehat{r}_j; \omega_c).$$

With $i_j = k'_j - (1-\delta_k)k_j$ and $y_j = z_j \varepsilon_j k_j^\alpha n_j^\nu$, the target is

$$\begin{aligned}\text{RHS}_j = \widehat{p}_j\left[y_j - \widehat{w}_j n_j - i_j - \text{AC}_j^k - \text{AC}_j^n(\widehat{w}_j)\right] \\ + \beta \mathbb{E}\left[\tilde{V}_{\text{nn}}(k'_j, n_j, \varepsilon'; Z', \Gamma_K(Z_j, K_j; \theta)) \,\middle|\, x_{f,j}, Z_j, K_j\right].\end{aligned}$$

The expectation uses the model's conditional shock transitions.[8] The next-period individual and aggregate states are explicitly $(k'_j, n_j, \varepsilon')$ and $(Z', \Gamma_K(Z_j, K_j; \theta))$. The value network minimizes

$$\mathcal{L}_{\text{val}}(\omega_V) = \frac{1}{B}\sum_{j=1}^{B}\left[\tilde{V}_{\text{nn}}(x_{f,j}; Z_j, K_j; \omega_V) - \text{RHS}_j\right]^2.$$

For fixed forecasting rules, training alternates between the conditional policy, discrete policy, and value network. Numerical settings are reported in Appendix B.1.

[8] For numerical stability, the value function on the right-hand side is evaluated using a target network.

Once training is complete, NN–KS proceeds as in the grid-based method. The trained networks are used to simulate the economy, and the simulated prices and aggregate capital are then used to update the forecasting rules. The networks are retrained under the updated forecasting rules, and this cycle of training, simulation, and forecasting-rule updates is repeated until convergence. NN–KS therefore changes the representation and numerical solution of the value and price-conditioned policy functions while retaining the economic problem and simulation structure of Sections 2 and 3.

4.2 Results

I evaluate NN–KS by comparing NN–KS with the KS benchmark. The computations were carried out on a system equipped with an AMD Ryzen 7 7700 CPU and an NVIDIA GeForce RTX 4060 Ti GPU. The results are mostly based on single precision (FP32). [9] It is important to note that the benchmark results for Bloom et al. (2018) reported here are based on a slightly modified version of their original code, with a modification to the way the distribution is updated during simulation. Consequently, these results may differ from those reported in the published version of Bloom et al. (2018). [10]

Table 2: Speed comparison for the Bloom et al. (2018) model

Method	Simulation Time (min)	Total Time (min, 20 outer loops)
KS	290	6141 ($\approx$ 4 days)
NN–KS	3	183

Note: Simulation time is measured for 5,000 periods during the first simulation. Total time covers 20 outer loops.

Computation and Simulation Speed. Table 2 presents the simulation and total computation times for each method. NN–KS reduces the total computation time from 6,141 minutes (approximately four days) to 183 minutes. A 5,000-period simulation falls from 290 minutes to 3 minutes.[11] The exact total time depends on the initial coefficients of the forecasting rules; the reported results is based on 20 outer-loop iterations using the original initial values.

Training in NN–KS proceeds cumulatively across the outer loops. Updating the forecasting rules changes the problem faced by the networks, but the change is gradual, so the networks obtained in the previous outer loop provide useful initial conditions for the next one. NN–KS

[9] The code is implemented primarily in single precision (FP32), with double precision used only for updating the distribution and aggregating variables in the simulation.

[10] In their original code, they set a threshold of 10^{-4} on the mass of firms in the distribution grid to determine which points to update. They ignore points with a mass below the threshold to reduce computation time. In the modified version, I update all points in the distribution grid regardless of their mass, and the results reported in this paper are based on this modified version. One could replicate the result by setting `disttol` = 0.0 and `GEerrorswitch` = 2 in their code.

[11] When the NN–KS simulation is run on the CPU alone, the same 5,000-period simulation takes approximately 38 minutes rather than 3 minutes on the GPU.

therefore does not relearn the solution from scratch in every outer loop. Instead, it progressively reduces the approximation error as the forecasting rules converge; after the first outer loop, approximately 3–4 minutes of additional training is typically sufficient before the next simulation.

For the present Bloom et al. (2018) application, however, the state space is still small enough for conventional grid-based value-function iteration. The accelerated grid-based method completes the same 20 outer loops in 108 minutes (Table 1), compared with 183 minutes for NN–KS. The grid-based solution is also more accurate in this case. As shown in Appendix B.2.1, its mean relative Bellman error is 3.69×10^{-7} at grid nodes and 5.45×10^{-6} at off-grid states, compared with approximately 3.22×10^{-4} for NN–KS in both samples. Thus, when a sufficiently fine grid remains feasible, the grid-based method is both faster and more accurate. The advantage of NN–KS emerges in larger models, where additional individual states, aggregate moments, or equilibrium prices make the storage and interpolation requirements of a grid-based policy table prohibitively large.

Simulation Paths and Business-Cycle Properties. Figure 5 reports unconditional simulation paths for aggregate variables over periods 2500–3000, computed using the policies from the final outer-loop iteration for both methods. Table 3 reports the corresponding HP-filtered business-cycle statistics.

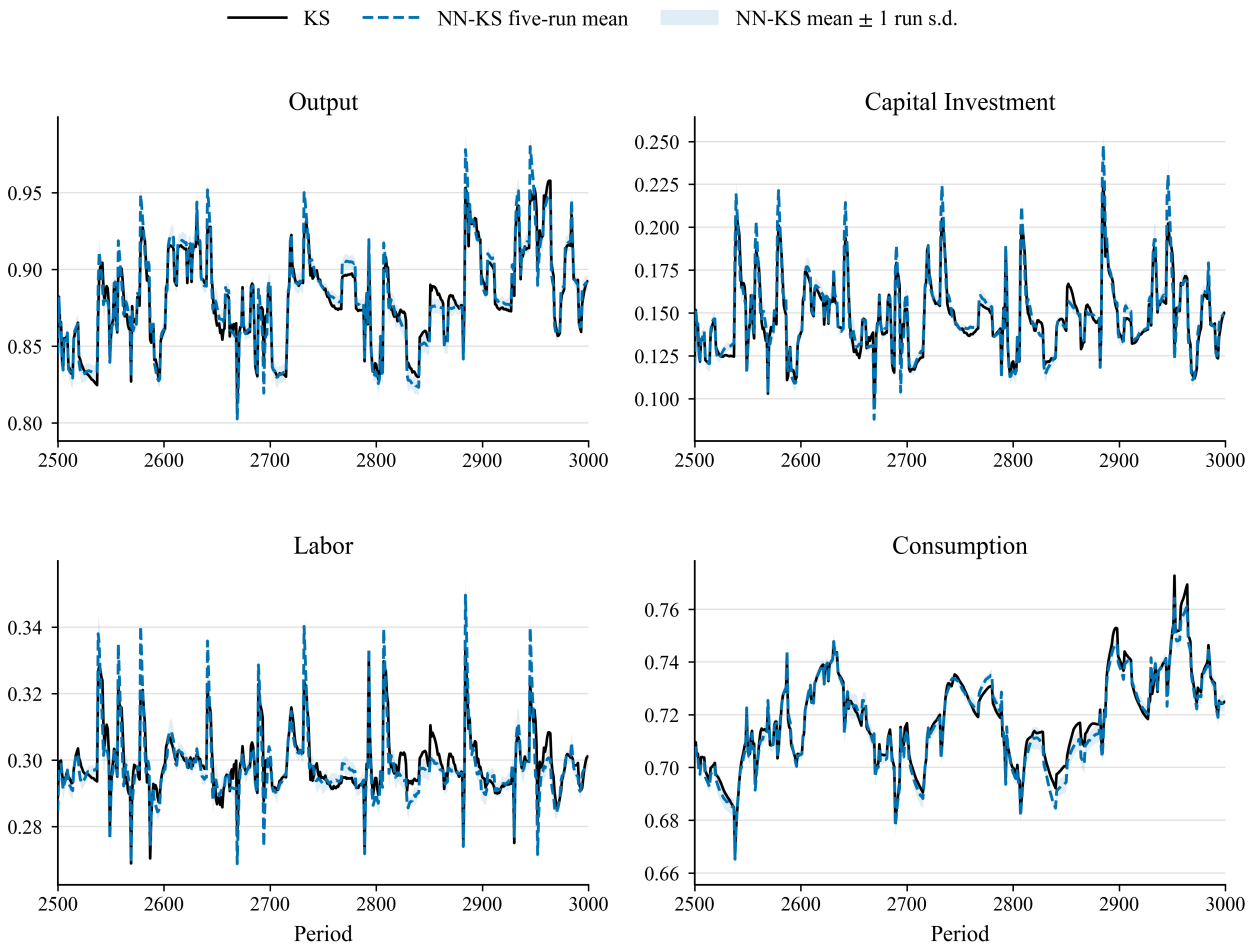
Table 3: HP-filtered business-cycle statistics

	Output	Investment	Labor	Consumption
Standard deviations (%)				
KS	1.8790	9.5546	2.1926	0.8673
NN–KS	2.0649	10.4607	2.5838	0.8836
(run s.d.)	0.0321	0.2580	0.0884	0.0177
Ratio to output				
KS	1.0000	5.0850	1.1669	0.4616
NN–KS	1.0000	5.0658	1.2512	0.4279
(run s.d.)	0.0000	0.0896	0.0379	0.0042
Correlations with output				
KS	1.0000	0.9440	0.7996	0.3303
NN–KS	1.0000	0.9521	0.8372	0.2844
(run s.d.)	0.0000	0.0031	0.0105	0.0439

Note: Statistics are based on the same realization of aggregate and stochastic volatility shocks, discarding the first 500 periods. Each series is expressed in logarithms, the HP filter with smoothing parameter $\lambda = 1600$ is applied, and the cyclical component is multiplied by 100 to express deviations in percent. NN–KS statistics are reported as averages over 5 independent runs. The third row in each block ("run s.d.") reports the across-run standard deviation computed from these 5 runs.

Figure 5 and Table 3 provide a consistent comparison. The NN–KS paths closely track the timing and overall movements of the KS benchmark, but the figure shows somewhat larger fluctuations in capital investment and labor. The business-cycle statistics quantify this difference: the standard deviation of capital investment rises from 9.5546 under KS to 10.4607 under NN–KS, while that of labor rises from 2.1926 to 2.5838. One possible explanation for this difference

Figure 5: Simulation paths generated by KS and NN–KS



Note: The figure reports simulation paths for periods 2500–3000 using the same realization of aggregate and stochastic volatility shocks. The solid black line represents KS, while the dashed blue line represents the mean of NN–KS across 5 independent runs evaluated at outer loop 20. The shaded area indicates plus or minus one across-run standard deviation. The panels report output, capital investment, labor, and consumption.

is that NN–KS allows continuous adjustment magnitudes, whereas the benchmark KS implementation restricts choices to a discrete grid without interpolating between candidate actions. This greater flexibility can make adjustment more attractive relative to inaction for firms near adjustment thresholds, potentially contributing to larger aggregate fluctuations. Nevertheless, the investment-to-output volatility ratio is nearly identical across the two methods, and the correlations of investment and labor with output remain close. Thus, NN–KS reproduces the broad simulation dynamics, although it somewhat overstates the volatility of capital investment and labor.

Table 4: Accuracy of the forecasting rules

	p (%)		K' (%)	
	KS	NN-KS	KS	NN-KS
Den Haan Statistics				
Maximum	3.80	3.59	5.68	5.77
Mean	0.96	0.82	1.83	1.81
Root Mean Squared Error				
RMSE	0.43	0.46	0.25	0.27
Forecast Regression R^2				
R^2	0.9793	0.9747	0.9957	0.9958

Note: The table reports Den Haan statistics (maximum and mean), root mean squared error (RMSE, %), and R^2 from forecast regressions, comparing the KS benchmark and the NN-KS method.

Accuracy of the Forecasting Rules. Table 4 presents the internal accuracy of the forecasting rules for the aggregate price p and next-period aggregate capital K'. I report the Den Haan statistics (maximum and mean), the root mean squared error (RMSE), and the R^2 from forecast regressions. The two methods produce comparable forecast accuracy. For p, NN–KS has lower maximum and mean Den Haan errors, a slightly higher RMSE, and a slightly lower R^2. For K', NN–KS has a slightly higher RMSE and a higher R^2, while its maximum Den Haan error is slightly higher and its mean error is slightly lower. Appendix B.2.2 provides detailed forecast accuracy statistics broken down by aggregate shock state for the Bloom et al. (2018) model application (see Table 10).

One possible explanation for the R^2 values below the approximately 0.99 typically reported in the literature is the (S, s)-type adjustment behavior of firms at distribution grid points with small probability masses. Retaining these grid points may make aggregate dynamics more difficult to forecast. The original implementation of Bloom et al. (2018) omits grid points with masses below 10^{-4} when updating the distribution. This truncation may help explain the higher R^2 of

approximately 0.99 reported in their study. [12] Extending the simulation to 10,000 periods yields only a slight improvement in forecast accuracy.

5 Household Preferences and Comovement after an Uncertainty Shock in Bloom et al. (2018)

This section examines how household preferences shape the general-equilibrium transmission of an uncertainty shock in Bloom et al. (2018). I first show how replacing the original infinite Frisch elasticity with a finite elasticity changes the wage response and generates comovement in output, consumption, investment, and employment. I then examine how consumption–hours complementarity affects the allocation between consumption and investment through the response of marginal utility.

5.1 The Wage Channel and Comovement

Original specification and the failure of comovement. An uncertainty shock raises the option value of postponing costly capital and labor adjustments. Firms therefore adopt a wait-and-see response: hiring, investment, and output decline even though aggregate productivity itself does not change. Whether consumption also declines depends on the balance between the losses in output and the resources released by lower investment and adjustment spending. The goods-market-clearing condition is

$$C_t = Y_t - I_t - AC_t, \tag{27}$$

where AC_t denotes aggregate capital and labor adjustment costs in units of goods. If investment and adjustment costs fall by more than output, consumption rises; if they fall by less, consumption declines. Thus, the failure of comovement in the original model is closely tied to the strength of the investment contraction.

Figure 6 illustrates this mechanism. Under the original specification, shown by the black line, output and employment fall on impact by approximately 3% and 5.5%, respectively, while investment falls by nearly 17%. The investment contraction is much larger than the output contraction, so consumption rises by about 1%. This is the source of the missing comovement in the original model.

To understand why investment falls so sharply, consider separable preferences with Frisch

[12] Applying the same truncation threshold can distort NN–KS simulations. Because NN–KS permits continuous choices, the distribution update distributes the mass associated with each choice across four neighboring grid points. In this implementation, the mass per grid point tends to be approximately one-quarter of that in the original implementation, making the same truncation threshold more restrictive.

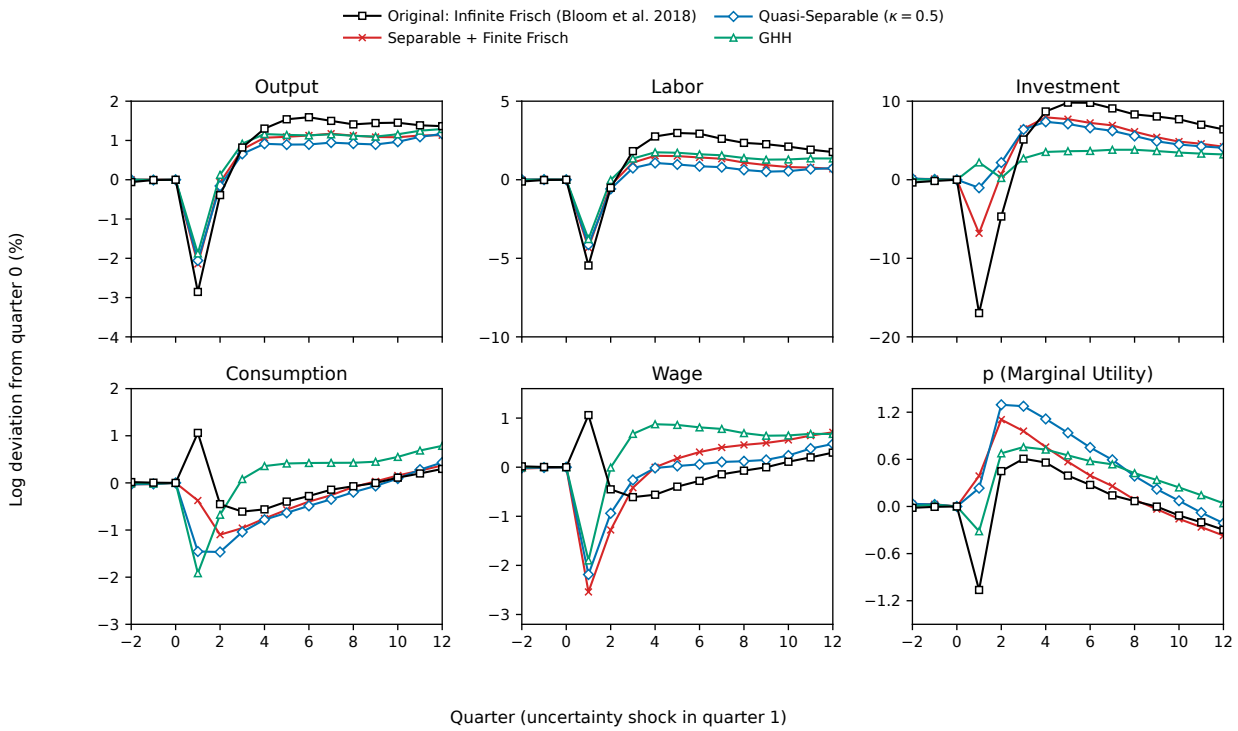


Figure 6: Uncertainty-shock responses under alternative household preferences

Notes: The uncertainty shock occurs in quarter 1. Responses are 100 times the log change in cross-economy mean levels from quarter 0. Each specification averages 2,000 economies computed with NN–KS. The paths are measured relative to their own pre-shock means. Table 5 reports the preference parameters.

elasticity $1/\phi$:

$$U(C,N) = \log C - \chi \frac{N^{1+\phi}}{1+\phi}, \tag{28}$$

$$p_t = U_C(C_t, N_t) = \frac{1}{C_t}, \qquad w_t = -\frac{U_N(C_t, N_t)}{U_C(C_t, N_t)} = \chi C_t N_t^{\phi}. \tag{29}$$

The original specification sets $\phi = 0$, implying an infinite Frisch elasticity and $w_t = \chi C_t$. Employment is absent from the wage equation, so the wage does not provide the usual equilibrating response to a fall in labor. As the black lines in the labor and wage panels of Figure 6 show, employment falls sharply while the wage rises by about 1%. The higher wage raises firms' wage bills and labor-adjustment costs and reinforces the decline in employment. Because production is Cobb–Douglas, capital and labor are complementary: at the firm level, the marginal product of capital is proportional to n_t^{ν}. The larger employment contraction therefore lowers the return to capital and makes investment less attractive, amplifying the investment decline. The fall in $p_t = 1/C_t$, shown in the bottom-right panel, partly supports investment by lowering the utility cost of current expenditure relative to future returns, but this force is not strong enough to offset the wage and employment channel.

Finite Frisch elasticity and restored comovement. With a finite Frisch elasticity, declining employment instead puts direct downward pressure on the wage:

$$\Delta \log w_t = \Delta \log C_t + \phi \Delta \log N_t. \tag{30}$$

The red line in Figure 6 uses a Frisch elasticity of 2. The wage now falls by 2.54% on impact rather than rising. This decline makes continued employment more attractive and limits the employment contraction to 4.33%. Preserving employment cushions the decline in the marginal product of capital, so investment falls by 6.81%, far less than under the original specification. The smaller release of resources from investment means that consumption falls by 0.37% as output declines by 2.14%; correspondingly, p_t rises by 0.39%. Output, consumption, investment, and employment therefore comove without any consumption–hours complementarity. Appendix C.2 shows that this result also holds for Frisch elasticities of 0.5 and 1.

5.2 Consumption–Hours Complementarity

Bloom et al. (2018) discuss GHH preferences as a possible way to change the consumption response. Under GHH preferences, consumption and hours are complements, so a decline in employment reduces the marginal utility of a given amount of consumption. This subsection examines whether introducing this channel causes consumption to fall with employment and thereby generates comovement after an uncertainty shock.

To vary complementarity while controlling the curvature of marginal utility, I use the quasi-separable preference family of Bilbiie et al. (2025):

$$U(C,N) = \frac{1}{1-\frac{\xi}{1-\gamma}} A(C,N)^{1-\frac{\xi}{1-\gamma}}, \qquad A(C,N) \equiv \frac{C^{1-\gamma}}{1-\gamma} - \chi \frac{N^{1+\phi}}{1+\phi}. \tag{31}$$

The induced marginal-utility price and real wage are

$$p_t = U_C(C_t, N_t) = C_t^{-\gamma} A(C_t, N_t)^{-\frac{\xi}{1-\gamma}}, \tag{32}$$

$$w_t = -\frac{U_N(C_t, N_t)}{U_C(C_t, N_t)} = \chi C_t^{\gamma} N_t^{\phi}. \tag{33}$$

Following Bilbiie et al. (2025), let ζ denote the local curvature of marginal utility, κ consumption–hours complementarity, and γ the wealth effect on labor supply. For equation (31),

$$\kappa(C,N) = \frac{\xi}{1-\gamma} \frac{C^{1-\gamma}}{A(C,N)}, \qquad \boxed{\zeta(C,N) = \kappa(C,N) + \gamma}. \tag{34}$$

Thus, holding ζ fixed while increasing complementarity requires a weaker wealth effect. Around

the reference allocation, the two price equations can be written as

$$\hat{p}_t \simeq -\bar{\zeta}\hat{c}_t + \bar{\kappa}\omega\hat{n}_t, \qquad \hat{w}_t \simeq \gamma\hat{c}_t + \phi\hat{n}_t, \qquad \omega \equiv \frac{\bar{w}\bar{N}}{\bar{C}}, \tag{35}$$

where hats denote log deviations and bars denote values at the reference allocation. A decline in consumption raises p_t, whereas a decline in hours lowers it when $\kappa > 0$. Through the wage equation, reducing γ also weakens the effect of consumption on labor supply.

I set $\bar{\zeta} = 1$, as under log-separable preferences, and compare $\bar{\kappa} = 0$, 0.5, and 1, with $\gamma = 1 - \bar{\kappa}$. The endpoints are separable and GHH preferences, while the intermediate case is quasi-separable. All three specifications use a Frisch elasticity of 2. Table 5 summarizes the calibration. Appendices D and D.1 provide the preference mapping and calibration details.

Table 5: Preference parameters in Figure 6

Specification	$\bar{\kappa}$	$\bar{\zeta}$	γ	Frisch elasticity
Original	0	1	1	∞
Separable	0	1	1	2
Quasi-separable	0.5	1	0.5	2
GHH	1	1	0	2

Notes: $\bar{\zeta}$, $\bar{\kappa}$, and the Frisch elasticity are evaluated at the reference allocation. In the quasi-separable family, ζ and κ can vary away from this allocation, while γ is constant.

The blue and green lines in Figure 6 show how this channel changes the finite-Frisch results. Output and employment still fall by roughly 2% and 4% on impact, but the consumption decline becomes larger as hours reduce marginal utility. Under the quasi-separable specification, consumption falls by 1.46%. Nevertheless, p_t rises by only 0.23%, because the decline in hours offsets much of the marginal-utility increase caused by lower consumption. Investment consequently falls by only 1.03%, so output, consumption, investment, and employment continue to comove.

At the GHH endpoint, consumption falls by 1.92%, but p_t falls by 0.32% and investment rises by 2.21%. Thus, the strong complementarity in GHH preferences successfully reverses the consumption response emphasized by Bloom et al. (2018), but it supports investment too strongly to generate full comovement in this calibration. The wage falls less under quasi-separable and GHH preferences than under separability, as shown in the wage panel, so the smaller investment contraction cannot be attributed to a larger wage decline. Instead, complementarity allows lower consumption to coexist with a smaller increase, or even a decline, in current marginal utility, reducing the utility cost of current investment relative to future returns. Appendix C.3 extends the comparison to five values of $\bar{\kappa}$.

6 Conclusion

This paper develops a method that substantially accelerates the Krusell–Smith method while preserving its familiar structure. The key idea is to precompute a price-conditioned policy function before simulation, so that each candidate price can be evaluated without resolving the firm's optimization problem. Expressing the candidate price as a deviation from its forecast also concentrates the approximation on the region relevant to the market-clearing search.

In the Bloom et al. (2018) model, which features fixed adjustment costs and discontinuous policies, computation time falls from 6,141 minutes to 108 minutes. To overcome the scalability limits of the grid-based approach, I extend the method using neural networks and a discrete–continuous representation of (S, s)-type policies. This NN–KS extension completes in 183 minutes and broadly reproduces the KS results. The same approach can be applied to other heterogeneous-agent models with implicit market-clearing prices, including models with multiple prices.

The main result of Section 5 is that a finite Frisch elasticity alone generates comovement after a pure uncertainty shock. It allows the real wage to fall with employment, cushioning the labor and investment contractions so that consumption also declines. Consumption–hours complementarity is not required for this result, but it further changes the allocation between consumption and investment by lowering marginal utility when hours fall.

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